# Econ 201B: Lecture 3

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Continuation of the proof for the one-stage deviation principle.

*Proof.* Let $s$ not be a BIE, then there is some player $i$ and history $h$ s.t. $i$ ha $s$ a profitable deviation $\hat{s}_i$ at history $h$. Choose a maximal such history (meaning there's no history with this property that has greater length), so

$$u_i\left(z_h\left(\hat{s}_i, s_{-i}\right)\right) > u_i\left(z_h\left(s_i, s_{-i}\right)\right)$$

Define a strategy $\bar{s}_i$

$$\bar{s}_i\left(\tilde{h}\right) = \begin{cases} s_i\left(\tilde{h}\right) & \forall \tilde{h} \in H_i, \tilde{h} \neq h \\ \hat{s}_i\left(\hat{h}\right) & \tilde{h} = h \end{cases}$$

We WTS that $\bar{s}_i$ is also a profitable deviation. Let $a = \hat{s}_i(h)$. Observe that

$$z_{ha}\left(\bar{s}_i, s_{-i}\right) = z_{ha}\left(s_i, -i\right)$$

while

$$u_i\left(z_{ha}\left(s_i, s_{-i}\right)\right) \geq u_i\left(z_{ha}\left(\hat{s}_{i,} s_{-i}\right)\right)$$

since $h$ was maximal and o.t.w would be strictly larger history which $\bar{s}_i$ is a profitable deviation, so

$$\begin{aligned} u_i\left(z_h\left(\bar{s}_i, s_{-i}\right)\right) &= u_i\left(z_{ha}\left(\bar{s}_i, s_{-i}\right)\right) = u_i\left(z_{ha}\left(s_i, s_{-i}\right)\right) \\ &\geq u_i\left(z_{ha}\left(\hat{s}, s_{-i}\right)\right) = u_i\left(z_h\left(\hat{s}_i, s_{-i}\right)\right) \\ &> u_i\left(z_h\left(s_i, s_{-i}\right)\right) \\ &\Rightarrow \\ u_i\left(z_h\left(\bar{s}_i, s_{-i}\right)\right) &> u_i\left(z_h\left(s_i, s_{-i}\right)\right) \end{aligned}$$

Thus, one state profitable deviations will yield a better payoff than the original strategy. □

## 1 Backwards Induction

Let $G = \left(N, A, H, P, \{\succsim_i\}_{i \in N}\right)$ be a PIG. $A(\emptyset)$ is the set of allowed initial actions for player $i = P(\emptyset)$. For each $b \in A(\emptyset)$, let $s^{G(b)}$ be some strategy profile for the subgame $G(b)$. Given some $a \in A(\emptyset)$, we denote $s^a$ the strategy profile for $G$ in which player $i = P(\emptyset)$ chooses the initial action $a$, and for each $b \in A(\emptyset)$ the subgame $G(b)$ is played according to $s^{G(b)}$. So, $s_i^a(\emptyset) = a$ and for every player $j$, $b \in A(\emptyset)$ and $bh \in H\backslash$, $s_j^a(bh) = s_j^{G(b)}(h)$

**Lemma 1.** *Backward Induction. Let $G = \left(N, A, H, P\{\succsim_i\}_{i \in N}\right)$ be a finite PIG. Assume that for each $b \in A(\emptyset)$ the subgame $G(b)$ has a BIE $s^{G(b)}$. Let $i = P(\emptyset)$ and let $a$ be the $u_i$-maximizing over $A(\emptyset)$. Then $s^a$ is a BIE of $G$.*

**Theorem 2.** *Every finite extensive form game with perfect information has a BIE.*

The punchline is that in (finite-horizon) PIGs the problem of strategic uncertainty has a straightforward solution: backward induction pins down successively the beliefs that each player has about the choices of those who move after the game.

# 2 Normal-Form Games

So far, we've only seen games written as PIGs but we can also write them in normal form.

**Definition 3.** A game in **normal form** (or strategic form) is a tuple $G = \left(N, \{S_i\}_{i \in N}, \{u_i\}_{i \in N}\right)$ where

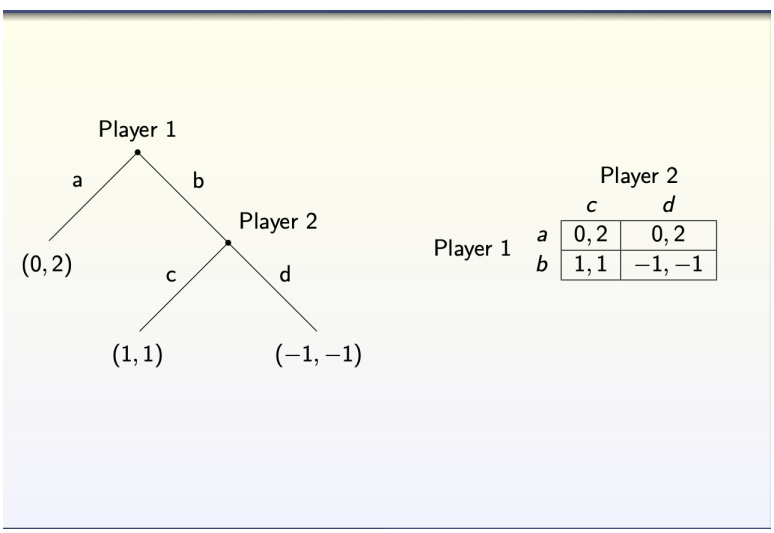
- $N$ is the set of players
- $S_i$ is the set of actions or strategies available to player $i$. We denote by $S = \prod_i S_i$ the set of strategy profiles.
- For each player $i \in N$, $u_i : S \to \mathbb{R}$ is a utility or payoff.

Consider the example where we rewrite a PIG as a normal form game. Let $G = \left(N, A, H, P, \{\succsim_i\}_{i \in N}\right)$ be a PIG and let $G'$ be the strategic form version given by $G' = \left(N', \{S_i\}_{i \in N}, \{u'_i\}_{i \in N}\right)$ where

- $N' = N$
- $S_i$ is the set of $G$-strategies of player $i$
- $u'_i(s) = u_i(z(s))$

Here is the example of the PIG from earlier written in normal form,

Figure 1 PIG vs Normal Form




## 2.1 Finite games

A finite game is a normal-form game in which the set of players $N$ is a finite set, and the set of strategy profiles $S$ is finite. We often adopt the convention that $N = [n]$.

The game of chicken is an example of a normal form game:

Table 1: Game of chicken

| P1\P2 | Go | Stop |
|---|---|---|
| Go | $(-100, -100)$ | $(5, -20)$ |
| Stop | $(-20, 5)$ | $(0, 0)$ |

Another example of a normal form is a Cournot oligopoly. In this game, there are $n$ firms where firm $i$ chooses a quantity $q_i \in \mathbb{R}_+$ and has a cost function $c_i : \mathbb{R}_+ \to \mathbb{R}$. The inverse demand is $d : \mathbb{R}_+ \to \mathbb{R}$. Firm $i$'s payoff is (profit) is

$$u_i(q_i, q_{-i}) = q_i d\left(\sum_{i=1}^{n} q_j\right) - c_i(q_i)$$

## 2.2 Strategy Profiles

Recall the following notational conventions from the handout. Fix a player $i \in N$. Represent a strategy profile as

$$s = (s_1, .., s_{i-1}, s_i, s_{i+1}, ..., s_n) = (s_i, s_{-i})$$

which means that given $s$, $s_{-i}$ is a profile of strategies for all players but $i$. This notation allows us to change strategy for $i$, say $s_i' \in S_i$, and compare $i$'s change in utility from $u_i(s) = u_i(s_i, s_{-i})$ to $u_i(s_i', s_{-i})$.

We can also write $S_{-i} = \prod_{j \neq i} S_j$ for the set of all profiles of strategies for players other $i$. So $s_{-i}$ is a typical element of $S_{-i}$. Thus, we can write

$$S = S_i \times S_{-i}$$

## 2.3 Mixed strategies

**Definition 4.** A **mixed strategy** for a player $i$ is a probability distribution $\sigma_i \in \Delta(S_i)$. Assume that $S_i$ is finite. Elements of $S_i$ are called pure strategies and pure strategies are degenerate mixed strategies.

If $\sigma = (\sigma_1, ...,_n)$ is a profile of mixed strategies, then we identify $\sigma$ with the probability distribution $\Delta(S)$ obtained from $(\sigma_i)_{i \in N}$ by using independence. We call $\sigma$ the independent mixture of $\sigma_1, ..., \sigma_n$. For example in the game of chicken if we had $\sigma_1(Stop) = 0.6$ and $\sigma_2(Go) = 0.5$ then $\sigma(Stop, Go) = 0.6 \cdot 0.5 = 0.3$.

Let $N = [n]$. The probability of $s \in S$ is $\sigma(s) = \times_{i=1}^{n} \sigma_i(s)$. Given $\sigma$, then, $i$'s expected payoff is

$$\begin{aligned} u_i(\sigma) = \mathbb{E}_\sigma[u_i(s)] &= \sum_{s_1 \in S_1} \sum_{s_2 \in S_2} ... \sum_{s_n \in S_n} \sigma_1(s_1) \cdots \sigma_n(s_n) u_i(s_1, s_2, ..., s_n) \\ &= \sum_{s \in S} \sigma(s) u_i(s) \end{aligned}$$

Consider the following example using the game of chicken

Table 2: Example

| P1\P2 | Go | Stop |
|---|---|---|
| Go | $(-100, -100)$ | $(5, -20)$ |
| Stop | $(-20, 5)$ | $(0, 0)$ |

The probability that agent 1 plays Go is $p$ and the probability that agent 2 plays Go is $q$. Therefore

$$u_i(\sigma) = -100pq + 5p(1-q) + 20(1-p)q + 0(1-p)(1-q)$$

This tells us the agent's overall expected payoff from playing the game conditional on player 2's strategies.

To summarize, we can represent the expected payoff of agent $i$

$$u_i(\sigma) = \mathbb{E}_{\sigma_i} u_i(s_i, \sigma_{-i}) = \sum_{s_i \in S_i} \sigma_i(s_i) u_i(s_i, \sigma_{-i})$$

where $u_i(s_i, \sigma_{-i})$ is itself the expected payoff of player $i$ given the probability distribution of the other players

$$u_i(s_i, \sigma_{-i}) = \mathbb{E}_{\sigma_{-i}} u_i(s_i, s_{-i}) = \sum_{s_{-i} \in S_{-i}} \sigma_{-i}(s_{-i}) u_i(s_i, s_{-i})$$

Similarly, we can represent it at

$$\begin{aligned} u_i(\sigma) = u_i(\sigma_i, \sigma_{-i}) &= \sum_{s_{-i} \in S_{-i}} \sigma_{-i}(s_{-i}) u_i(\sigma_i, s_{-i}) \\ &= \sum_{s \in S} \sigma_i(s_i) \sigma_{-i}(s_{-i}) u_i(s_i, s_{-i}) \end{aligned}$$

# 3 Dominated Strategies and Rationalizability

We've yet to discuss what strategies agents *should* in a normal form game. Consider a normal-form game $G = (N, \{S_i\}, \{u_i\})$. Recall the problem of strategic uncertainty: Player $i$'s optimal choice depends on what $i$ believes other players will do. In some special games, player $i$ has an optimal choice regardless of what $i$ believes. In other special games, $i$ can use other players' rationality to uniquely predict what they will do. In many more games, $i$ can use other players' rationality, and what other players infer about others' rationality, to narrow down what $i$'s opponents will do.

## 3.1 Dominant strategies

Let $(N, \{S_i\}, \{u_i\})$ be a finite normal-form game, with $N = [n]$.

**Definition 5.** A strategy $\bar{\sigma}_i \in \Delta(S_i)$ is a **dominant strategy** for player $i$ if

$$u_i(\bar{\sigma}_i, \sigma_{-i}) > u_i(\tilde{\sigma}_i, \sigma_{-i})$$

for all $\tilde{\sigma}_i \in \Delta(S_i)$ with $\tilde{\sigma}_i \neq \tilde{\sigma}_i$ and all $\sigma_{-i} \in \times_{j \neq i} \Delta(S_j)$.

### 3.1.1 Prisoner's Dilemma

Classic game where each agent can either choose to cooperate with each other by lying to the police or defect and confess. The bi-matrix of the normal form game is given by

Table 3: Prisoner's Dilemma

| P1\P2 | Confess | Cooperate |
|---|---|---|
| Confess | $(-3, -3)$ | $(0, -10)$ |
| Cooperate | $(-10, 0)$ | $(-2, -2)$ |

In this special game, confess is the dominant strategy, so they will always confess *regardless* of what the other agent does.

## 3.2 Dominated strategies

**Definition 6.** A strategy $\sigma_i \in \Delta(S_i)$ is a **dominated strategy** for player $i$ if there exists a strategy $\bar{\sigma}_i \in \Delta(S_i)$ such that

$$u_i(\bar{\sigma}_i, \sigma_{-i}) > u_i(\sigma_i, \sigma_{-i})$$

for all $\sigma_{-i} \in \times_{j \neq i} \Delta(S_j)$.

We say that $\bar{\sigma}_i$ dominates $\sigma_i$.

**Lemma 7.** *A dominant strategy is always pure.*

**Lemma 8.** *A strategy $\bar{\sigma}_i$ dominates $\sigma_i$ iff $u_i(\bar{\sigma}_i, s_{-i}) > u_i(\sigma, s_{-i})$, for all pure strategy profiles $s_{-i} \in S_{-i}$.*

Table 4: Normal form game

| P1\P2 | $L$ | $R$ |
|---|---|---|
| $U$ | $(2,1)$ | $(0,0)$ |
| $M$ | $(0,1)$ | $(2,0)$ |
| $D$ | $(0,0)$ | $(0,1)$ |

## 3.3 Best Response

Consider the following normal form game

In this example, $D$ is dominated and we can conclude that P1 does not play $D$. P2 will realize this, and given that they conclude that $R$ is dominated by $L$ and will not play $R$. P1 will realize this, and given that will conclude that $M$ is dominated by $R$. Thus, P2 must play $L$ and P1 will play $U$.

**Definition 9.** A **belief** for player $i$ is a probability distribution $\mu \in \Delta(S_{-i})$.

**Definition 10.** A strategy $\sigma_i \in \Delta(S_i)$ is a **best response** to beliefs $\mu$ if it solves the problem of maximizing $\mathbb{E}_\mu[u_i(\sigma_i', \tilde{s}_{-i})]$ over $\sigma_i' \in \Delta(S_i)$.

We will denote the set of best responses to beliefs $\mu$ by $\beta_i(\mu)$. Note that $\beta_i(\mu)$ is always non-empty in finite games, and it may not be a singleton.

A mixed strategy $\sigma_i$ is in $\beta_i(\mu)$ iff every pure strategy in the support of $\sigma_i$ is in $\beta_i(\mu)$. In particular, every strategy in the support of $\sigma_i$ yields the same payoff to $i$.

## 3.4 Domination Theorem Rephrased

**Theorem 11.** *In a finite game, a strategy is dominated iff there is no belief to which it is a best response.*